

PILSEO PARK

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RESEARCH INTEREST

Face Modeling, Digital Human, Physics Simulation, Neural Rendering, Computer Graphics, 3D/4D Vision

EDUCATION

- Ph.D. in Computer Science** at Michigan State University 08/2023 – Present
Advisor: Prof. Yiyong Tong
- M.E. in AI Application** at Chung-Ang University 09/2020 – 06/2022
Advisor: Prof. Youngbin Kim
Dissertation: Cross-Modal Semantic Alignments Module for Image-Text Retrieval
- B.S. in Computer Science** at Oklahoma State University 08/2015 – 05/2019
Cum Laude

EXPERIENCE

- Michigan State University** – *Research Assistant* 08/2023 – Present
- Collaborated with Qualcomm to develop an efficient Gaussian allocation method for high-fidelity 3D head reconstruction; accepted for publication at *WACV* 2027.
 - Developed a generalizable framework for reconstructing animatable 3D Gaussian head avatars from a single image using geometry image and symmetry-guided completion; accepted for publication at *ACCV* 2026.
 - Developing a physics-guided framework for human–object reconstruction, focusing on reducing artifacts around hand–object contact and improving computational efficiency.
- 4by4 Inc.** – *AI Engineer* 08/2022 – 06/2023
- Developed a dual-color-space network leveraging global priors from RGB and YCbCr representations for natural and realistic photo retouching; published in *Scientific Reports* (2023).
- Chung-Ang University** – *Research Assistant* 09/2020 – 06/2022
- Developed a graph-based cross-modal semantic alignment module that explicitly models relationships between image regions and textual tokens for image–text retrieval; published in *Multimedia Tools and Applications* (2024).

PUBLICATION

*: equal contribution

Pilseo Park*, Andrew Hou*, Michel Sarkis, Ning Bi, Yiyong Tong, and Xiaoming Liu. “EGGHead: Efficient Generalizable Gaussian Heads from a Single Image,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027. **(Round 1)**

Pilseo Park, Fizza Rubab, and Yiyong Tong. “Learning Semantic Inpainting for Animatable Gaussian Head Avatars.” *Asian Conference on Computer Vision (ACCV)*, 2026.

Pilseo Park, Ze Zhang, Michel Sarkis, Ning Bi, Xiaoming Liu, and Yiyong Tong. “Refined Geometry-guided Head Avatar Reconstruction from Monocular RGB Video,” *arXiv*, 2025. [PDF](#)

Pilseo Park, Soojin Jang, Yunsung Cho, and Youngbin Kim. “SAM: Cross-Modal Semantic Alignments Module for Image-Text Retrieval,” *Multimedia Tools and Applications*, 2024. [PDF](#)

Pilseo Park, Heungmin Oh, and Hyuncheol Kim. “Dual-Color Space Network with Global Priors for Photo Retouching,” *Scientific Reports*, 2023. [PDF](#)

ACADEMIC SERVICES

Teaching Assistant at Michigan State University: CSE 472 Computer Graphics; CSE 471 Media Processing and Multimedia

SKILLS AND INTERESTS

Programming	Python, C/C++, C#, Java, HTML, PHP, OpenGL, MATLAB
Framework	PyTorch, TensorFlow